

■ SDG 4 – QUALITY EDUCATION

Mentora AI

AI-Powered Personalised Learning Assistant s Mentor Platform

Lenovo LEAP Generative AI s Agentic Systems Engineering Internship Program

Presented by **Pritesh Patro**

The Problem

Today's students face persistent learning gaps that traditional classrooms simply cannot close.

No Personalised Tutoring

Every student learns differently, yet most receive a one-size-fits-all education.

Teacher Overload

One teacher cannot focus on every individual student simultaneously.

Unresolved Doubts

Difficult concepts remain unclear with no instant doubt-solving mechanism.

Late Identification

Teachers cannot identify struggling students until it is too late.

Passive Platforms

Existing learning tools deliver content but lack interactive, adaptive intelligence.

■ THE SOLUTION

Introducing Mentora AI

An Agentic AI platform that delivers personalised learning whilst empowering mentors to identify and support struggling students — in real time.

AI Tutor

AI Summariser

Quiz Generator

Flashcards

Concept Maps

Voice Quiz

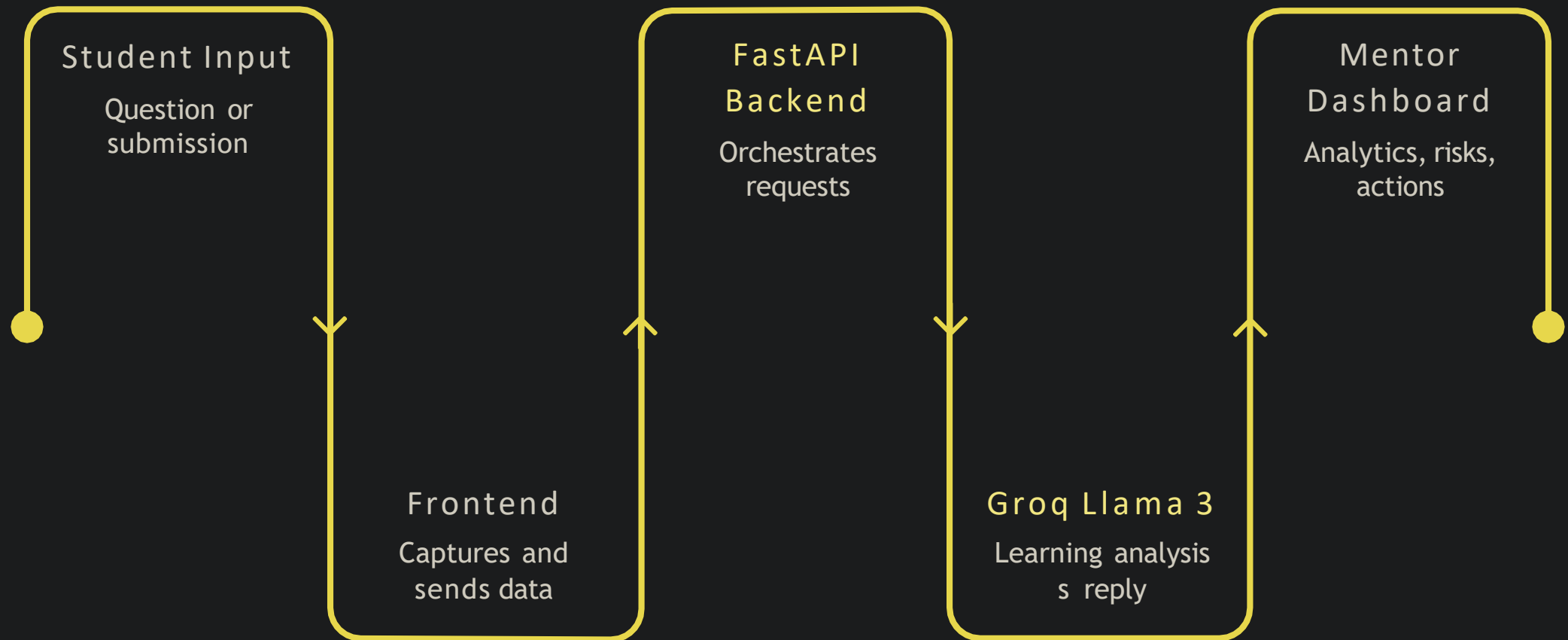
ELI5 Learning

Feynman Checker

Mentor Dashboard

Dropout Risk Prediction

How It Works



From the moment a student asks a question, Mentora's agentic pipeline processes input through a FastAPI backend, leverages Groq's Llama 3 for intelligent analysis, and surfaces actionable insights on the mentor dashboard — all backed by a lightweight SQLite database.

■ TECHNOLOGY STACK

Built on a Modern Stack



Frontend

HTML · CSS · JavaScript



Backend

FastAPI · Python



Artificial Intelligence

Groq Llama 3



Database

SQLite



Deployment

Render



Version Control

GitHub

Features That Empower

AI Tutor

Conversational learning
assistant

Smart Notes

Summariser

Condense lengthy
material instantly

Flashcards

Active recall for retention

Quiz Generator

Auto-created
assessments

Concept Maps

Visualise knowledge
connections

Voice Quiz

Hands-free spoken
testing

ELI5 Mode

Explain it like I'm five

Feynman Checker

Verify true understanding

Mentor Dashboard

Real-time student
monitoring

Student Analytics

Performance trend
insights

Dropout Prediction

Early risk identification

Responsive

Interface

Seamless across all
devices

Real World Impact

24/7

Learning Support

Always-on AI assistance



Student Performance

Measurable improvement

Early

Risk Detection

Identify struggling students

AI

Personalised Guidance

Adaptive to every learner



Teacher Workload

Automated insights save time

SDG 4

Quality Education

Advancing global goals

Future Scope

The next evolution of Mentora AI extends beyond the web — bringing intelligent learning to every device, language, and classroom.

Mobile App

iOS & Android native experience

WhatsApp AI Tutor

Learning through familiar chat

PDF Chat

Converse with any document

OCR Notes Scanner

Digitise handwritten notes

Attendance Prediction

Forecast student engagement

Emotion Recognition

Detect frustration & confusion

Regional Languages

Multi-lingual AI support

Cloud Database

Scalable institutional storage

Institution Dashboard

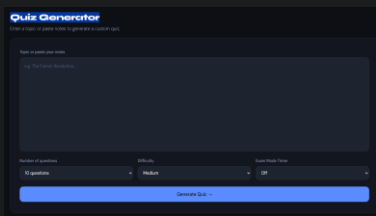
School-wide analytics overview

Analytics Reports

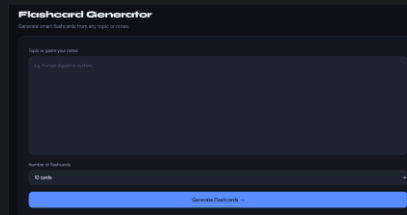
Exportable performance insights

See Mentora in Action

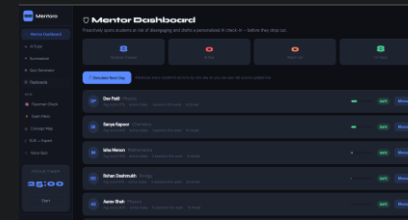
Experience the full platform deployed live on Render. Explore the student journey from login to analytics, and the mentor's command centre for monitoring at-risk learners.



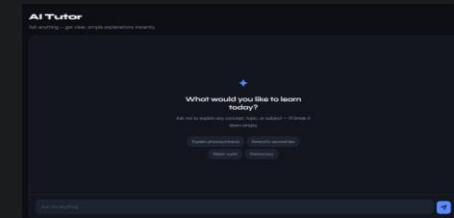
Quiz Generator



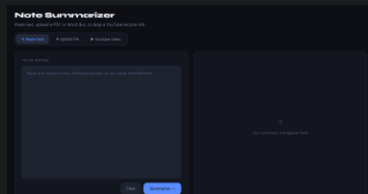
Flashcards



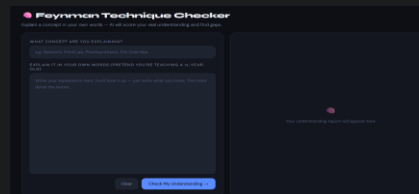
Mentor Dashboard



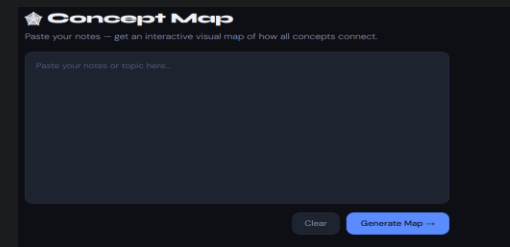
AI Tutor



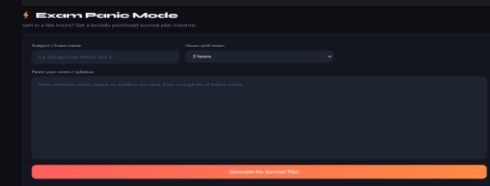
Summariser



Feynman



Concept Map



Exam Panic Mode

■ THANK YOU

Thank You

Learn Smarter. Teach Better. Empower Every Student.

Project

Mentora AI

GitHub Repository:

[https://github.com/
Pritesh-
05/mentora-ai](https://github.com/Pritesh-05/mentora-ai)

Render

Deployment:

[https://mentora-ai-
jyvi.onrender.com/](https://mentora-ai-jyvi.onrender.com/)

Powered By

Groq AI · FastAPI · SQLite

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